6TH GRADE MATH · UNIT 4: EXPRESSIONS AND EQUATIONS

Arcade Party Planner

Write expressions and solve equations to plan a birthday party at the arcade.

Honors

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Your cousin wants a birthday party at Pixel Palace Arcade. Each guest needs a game card, the party room costs a flat fee, and prizes are bought with tickets.

You will write expressions and solve equations to compare party packages and plan a party that fits the budget.



Party Room

$40 flat fee

up to 3 hours

Game Card

$12 per guest

1 credit = 1 game

Learning Goals

- I can write and interpret multi-term expressions, including exponents.
- I can generate equivalent expressions and justify them with properties.
- I can use equations to compare plans and find break-even points.

Project Timeline

When	What you will do
Day 1	Model the Packages
Day 2	Tickets & Powers
Day 3	Optimize the Party
Final	Party Package Proposal + reflection

Part 1: Model the Packages

Day 1 · Use the Arcade Price Board. g = guests.

Arcade Price Board

Game Card	Pizza	Drinks
$12 per guest	$15 per pizza	$2 per guest
60 credits	serves 4	

Party Room	Prize Tickets	Cake
$40 flat fee	Ticket bonus: 50	$25 per cake
	per guest	

1. Write the cost of the regular package (room, game card, drink, and pizza at $\frac{1}{4}$ pizza per guest) and simplify.

2. The Mega costs 20g. For what number of guests do the two packages cost the same? (Solve by testing values or by reasoning.)

3. A coupon gives $3 off each game card. Write two equivalent expressions for the new regular cost and explain each.

4. Evaluate both of your coupon expressions for g = 16 to show they are equivalent.

Part 2: Tickets & Powers

Day 2 · Prize prices are written with exponents.

5. Order these prizes by ticket cost: 2^7, 5^3, 11^2, 3^5.

6. Each guest wins an average of $2(t + 15)$ tickets where t is bonus tickets. Rewrite without parentheses and find the total for 12 guests with $t = 50$.

7. Factor $48g + 36$ and $45g + 30g^2$, and explain what the factored forms show.

8. Is $3(g + 4)$ equivalent to $3g + 4$? Use a value of g and a property to explain.

Part 3: Optimize the Party

Day 3 · The family budget is $320.

9. How many guests can attend each package within $320? (Use equations, then round sensibly.)

10. With a $3-off coupon, recompute for the regular package. Which package now allows more guests?

11. Design a package for Pixel Palace that is cheaper than the Mega for 10 guests but costs more than the regular package for 25 guests. Write its expression and show both comparisons.

Final Product: Party Package Proposal

Your final product must include:

- Simplified expressions for every package
- Break-even analysis with a table or graph
- Equivalent expressions justified with properties
- Budget equations and solutions
- Your designed package with both comparisons shown

Reflection

R1. Explain the break-even point in words a parent would understand.

R2. Why must you round down guest counts when solving budget equations?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Expressions and equations model every cost correctly and are used to compare packages and plan within the budget. Goes beyond the minimum with a creative, realistic choice.	Expressions and equations model every cost correctly and are used to compare packages and plan within the budget.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes expressions, a comparison table, solved equations, and a clear recommendation. Polished, organized, and visually engaging.	Proposal includes expressions, a comparison table, solved equations, and a clear recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.